

The MACD generates several types of signals:

- **MACD line crossing above signal line:** Bullish signal
- **MACD line crossing below signal line:** Bearish signal
- **MACD line crossing above zero:** Strengthening uptrend
- **MACD line crossing below zero:** Strengthening downtrend
- **Divergence between MACD and price:** Potential trend reversal

The MACD is particularly valued for its versatility—it provides information about both trend direction and momentum in a single indicator.

Stochastic Oscillator: Measuring Price Momentum

Developed by George Lane in the 1950s, the Stochastic Oscillator compares a security's closing price to its price range over a specific period. Like the RSI, it oscillates between 0 and 100 but uses a different calculation method.

The concept behind the Stochastic is that during uptrends, closing prices tend to be near the high of the recent range, while during downtrends, they tend to be near the low of the recent range. It's like observing whether a runner finishes near the front or back of the pack to gauge their competitive strength.

```
1. def calculate_stochastic(data, k_period=14, d_period=3):
2.     # Calculate %K (fast stochastic)
3.     low_min = data['Low'].rolling(window=k_period).min()
4.     high_max = data['High'].rolling(window=k_period).max()
5.
6.     k = 100 * ((data['Close'] - low_min) / (high_max - low_min))
```